

Problem 1: Creating a Dataset of Patients

Background: You are analyzing 10 patients' cardiovascular profiles. The dataset includes **Age**, **Gender**, **Blood Pressure**, **Cholesterol**, and **Diagnosis**.

Task:

Recreate the `patients_data` data frame exactly as in the lecture and use `View()` to inspect it.

Problem 2: Select Key Clinical Columns

Background:

You want to prepare a report containing only Name, Age, Cholesterol, and Diagnosis.

Task:

Use `select()` to extract those columns from `patients_data`.

Problem 3: Filter Patients by Age and Gender

Background:

You need to identify **female patients over 35 years old** for a follow-up consultation.

Task:

Use `filter()` to extract rows where `Gender == "F"` and `Age > 35`.

Problem 4: Identify Hypertensive Males

Background:

You want to find all **male** patients diagnosed with **Hypertension** for a cardiovascular drug trial.

Task:

Filter the dataset for `Gender == "M"` and `Diagnosis == "Hypertension"`.

Problem 5: Select Borderline Female Patients

Background:

The clinical nutrition team needs to target **borderline female patients** for lifestyle counseling.

Task:

Use `filter()` to extract rows where `Gender == "F"` and `Diagnosis == "Borderline"`.

Problem 6: Arrange by Cholesterol Level

Background:

You are preparing a ranked list of patients by cholesterol for a lipid management program.

Task:

Use `arrange()` to display patients from **lowest to highest** cholesterol. Then arrange again in **descending** order.

Problem 7: Create a Combined Blood Pressure Score

Background:

To assess total cardiovascular load, physicians often use **BP_Sys + BP_Dia** as a combined indicator.

Task:

Use `mutate()` to create a new column called `"BP_Sum" = BP_Sys + BP_Dia`.

Problem 8: Calculate Average Age and Cholesterol

Background:

The hospital's quality assurance department wants summary statistics.

Task:

Use `summarise()` to calculate:

- Mean age (`mean_age`)
- Mean cholesterol (`mean_chol`)

Problem 9: Compare Cholesterol by Diagnosis Group**Background:**

You suspect cholesterol differs across diagnostic groups.

Task:

Use `group_by(Diagnosis)` with `summarise()` to calculate mean age, mean cholesterol, and patient count (`n`) per group.

Problem 10: Identify Highest Risk Group**Background:**

After grouping, determine which diagnosis category has the **highest average cholesterol**.

Task:

Use `arrange(desc(mean_chol))` after your summarization to rank the groups.

Problem 11: Analyze Female Subgroup**Background:**

A female-only lipid profile study needs to be generated.

Task:

Use `filter(Gender == "F") → group_by(Diagnosis) → summarise(mean_age, mean_chol, n)` and arrange by `mean_chol`.

Problem 12: Remove Derived Variables**Background:**

You mistakenly added extra columns during analysis. Before exporting the data, you must remove derived ones.

Task:

Use `select()` to keep only the original columns (`ID`, `Name`, `Age`, `Gender`, `BP_Sys`, `BP_Dia`, `Cholesterol`, `Diagnosis`).

Problem 13: Convert Blood Pressure Data to Long Format**Background:**

You want to plot systolic and diastolic BP trends in one column for easy visualization.

Task:

Use `pivot_longer()` on `BP_Sys` and `BP_Dia` to create columns:

- `"BP_Type"`
- `"Value"`

Problem 14: Convert the Long Format Back to Wide Format**Background:**

After merging your reshaped dataset with another file, you need to return to the original wide form.

Task:

Use `pivot_wider()` to recreate separate columns for `"BP_Sys"` and `"BP_Dia"`.

Problem 15: Compare Two Study Cohorts (Medium–Hard)

Background:

Two separate cardiovascular studies collected patient IDs.

Cohort A (current study):

c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)

Cohort B (follow-up study):

c(2, 3, 7, 10, 11, 12)

Task:

1. Find which patients are **common to both cohorts** using `intersect()`.
2. Find patients **unique to Cohort A** using `setdiff()`.
3. Print both results and briefly describe what each represents in a real-world biomedical context.